Worldbuilding Test Report

Local regression and D01 verification | 29 October 2026 | Asia/Hong_Kong

1. Executive summary

The latest main branch was pulled successfully. This run tests commit 55d059e70ee66b96f0093737bb23f503fa3ed3ec together with the uncommitted login-label contrast fix and its browser regression assertion. Registration validation and the Windows test-runner adjustment are already committed. The tested working tree is identified by non-secret source hashes in the evidence package.

Verification layer	Current result
Backend leaf cases	78 total: 75 passed, 0 failed, 3 skipped
Browser workflows and local read observation	13 passed, 0 failed, 0 skipped
Real Gemini smoke retest	3 real-provider scenarios passed; 2 repeated fixture cases also passed
Frontend unit tests	2 passed
Lint, TypeScript and production build	All 3 checks passed
Backend coverage	96.46% lines; 89.09% branches; 98.03% functions
D01 malformed registration email	Fixed and verified through real HTTP/MySQL test Q14
D02 login label contrast	Fixed and verified: approximately 5.57:1 against 4.5:1

Both previously reported findings are resolved in this tested working tree. The results support the exercised local workflows and safeguards. Remaining work includes repeated AI quality evaluation, broader accessibility, hosted performance and client acceptance testing.

What changed since 26 October

Registration now shares the profile endpoint's email-format rule and rejects malformed addresses before database access. Six additional regression cases cover account flows and limits. The login label color is now #7a5e24, and UI12 explicitly checks for regression of its contrast failure. Historical reports and evidence are preserved separately.

Test scope and method

2. Environment and isolation

Execution used Windows, Node v24.19.0, MySQL 8.4.9 (MySQL Community Server - GPL), Playwright 1.56.1 and c8 10.1.3. Browser automation used Chromium 141.0.7390.37 at 1440 x 1000, with a 390 x 844 login observation.

A separate MySQL instance on the loopback interface used a dedicated temporary data directory. Integration and browser tests created synthetic databases and accounts. No test writes targeted the normal application database. After the run, verification found zero remaining matching temporary schemas. Windows terminated the browser server without its cleanup handler, so the exact fixture database was explicitly removed and the cleanup query was repeated successfully.

3. Strategy and case selection

Technique	Executed example	Expected property
Equivalence partitions	Guest, valid/expired session, roles and outsider	Only permitted operations succeed
Boundary values	Username 50/51; email 100/101; body input 5999/6000/6001	Accept supported limits and reject excess
State transitions	Draft, submit, approve/reject, withdraw and restore	Reject stale/repeated invalid transitions
Failure injection	Database/provider substitutes and browser request failures	Preserve input and avoid unintended publication
Real integration	HTTP handlers with isolated SQL; separate Gemini calls	Verify persistence and provider contracts

Controlled AI responses make browser recovery tests repeatable. Live-provider trials run separately and do not contribute to deterministic backend coverage. Failure-injection tests using a substitute do not establish recovery from a real database outage.

D01 acceptance criterion

POST /register with email not-an-email must return HTTP 400 and create no account. Q14 passed against the real test database. The additional registration suite verifies rejection before any database query, then checks successful registration, hashed passwords, duplicate rejection and login using the actual HTTP handlers with a database substitute.

Workflow verification

4. Browser scenarios

ID	Scenario	Result
N01	60 local reads at concurrency five	PASS
UI01	Registration mismatch, success and duplicate rejection	PASS
UI02	Login failure, keyboard password toggle and profile update	PASS
UI03	Guest/reader controls, private exclusion and reverse relationship	PASS
UI04	Author submission and Owner approval publish exact content	PASS
UI05	Stale proposal preserves input and supports explicit recovery	PASS
UI06	Same-tab refresh restores unsaved input without publishing	PASS
UI07	History rendering and restore API retain earlier versions	PASS
UI08	Missing AI configuration disables composer safely	PASS
UI09	Mocked AI diff, dismiss, apply, stale protection and save	PASS
UI10	Draft snapshot refresh and mocked AI failure/retry	PASS
UI11	Revoked membership retains copyable local input	PASS
UI12	Search results and mobile login accessibility observation	PASS

UI07 restores through an HTTP request after opening history. It does not fully automate every compare/restore button. UI09 and UI10 use clearly mocked AI replies with actual application persistence. UI12 includes an explicit regression assertion for the login-label contrast finding. The final sampled-page scan reports zero violations; this is not a whole-application audit.

Acceptance traceability

Account tests Q01-Q08 and Q13-Q14 cover authentication and input handling. Q09-Q12 and the legacy permissions suite exercise visibility, search, transactional rollback and conflict recovery. UI04-UI07 connect those backend state checks to collaboration and history flows. AI unit/API tests and UI08-UI11 cover controlled suggestions and recovery. These are workflow themes rather than an approved complete User Story denominator.

Execution and coverage

5. Counts and interpretation

The backend JUnit file contains 78 leaf cases: 75 pass and three skip. Node's TAP summary includes three parent containers and therefore reports 81 tests: 78 pass and three skip. There are no failed backend cases. The live run repeats two controlled fixtures and executes the three previously skipped Gemini scenarios; those repeats are not extra unique cases.

Metric	Covered / total	Coverage
Lines	1174 / 1217	96.46%
Branches	735 / 825	89.09%
Functions	50 / 51	98.03%

Backend runtime file	Lines	Branches
ai-config.js	100%	100%
assistant-api.js	100%	100%
assistant-connections.js	100%	87.5%
assistant-edit.js	100%	97.14%
assistant-prompts.js	100%	100%
assistant-router.js	93.75%	88.88%
assistant.js	98.94%	90.66%
content-api.js	97.65%	83.95%
db.js	89.28%	50%
gemini.js	100%	96.61%
proposal-api.js	100%	85.42%
relationships.js	94.91%	90.9%
security.js	100%	91.3%
server.js	89.82%	89.02%

c8 includes all 14 top-level backend runtime JavaScript files. Dependencies, test/diagnostic scripts, SQL and frontend source are excluded. Coverage includes paths exercised with substitutes. Line and statement totals share V8-derived instrumentation, and compact source lines limit interpretation. High coverage alone cannot establish requirement completeness, semantic correctness or production readiness.

AI, accessibility and load

6. Real Gemini retest

Scenario	Result	Elapsed
real Gemini HTTP smoke with synthetic context	PASS	3.29 s
real Gemini body edit produces a complete validated replacement	PASS	3.38 s
unsaved draft consistency check with a selected published reference	PASS	5.16 s

The separate live-provider trials ran earlier on 29 October and remain applicable to the unchanged AI runtime code. The configured model was gemini-3.8-flash. Conversation returned the daylight-only entry rule. Consistency checking cited Harbour Regulations, ID 3 version 2, identified midnight entry as a conflict and suggested waiting until dawn. The body-edit trial passed the complete replacement contract, but its full replacement text is not logged for independent semantic review.

Each live scenario ran once using synthetic content and controlled context. The measurements include application/test overhead and are not a latency guarantee. A repeated five-scenario human-rubric evaluation remains pending. The real replies and runner outcomes are retained in live-ai-20261029.tap and its JUnit file.

7. Resolved finding D02

The earlier mobile login scan found 4.45:1 contrast on .step-label, below the required 4.5:1. The shared label color changed from #8b6d2f to #7a5e24 against #f8f5ec, giving approximately 5.57:1. The final 390 x 844 axe scan reports zero violations, and UI12 now asserts no contrast failure for that label. The complete browser suite and frontend build checks passed after the CSS change. This sampled scan does not establish whole-application WCAG conformance.

8. Local read observation

The backend served 60 reads at concurrency 5 against a two-entity synthetic fixture, with 0 non-200 responses. Elapsed time was 119.9 ms, p50 7.14 ms and p95 31.32 ms. There was no sustained load, large dataset, AI workload or agreed acceptance threshold. These local observations do not establish hosting capacity.

At the sampled mobile login viewport, document and viewport widths were both 390px, with no horizontal overflow.

Remaining work and evidence

9. Priorities

1. Retain the passing registration and contrast regression checks. 2. Repeat AI scenarios covering true conflicts, intentional revisions, insufficient evidence, hostile retrieved instructions and fact-preserving rewrites. 3. Expand device/browser and keyboard coverage, relationship editing, and full version-control UI interactions. 4. Verify representative data volumes, hosted configuration, latency/load targets, outages and security controls. 5. Map the official User Story backlog and record client UAT.

Not included: exhaustive security testing, frontend source coverage, long-duration stress/availability, production hosting verification, real simultaneous browser co-editing, automatic merge or proof of general AI semantic accuracy.

10. Evidence package

Artifact	Purpose
test-case-results.csv	78 backend outcomes, 13 browser outcomes and 3 separately listed live retests
evidence/backend.tap and backend.junit.xml	Raw deterministic runner and leaf outcomes, including Q14
coverage/index.html and coverage-summary.json	Inspectable file/line coverage and machine totals
evidence/browser-results.json; screenshots/	Actual browser outcomes and representative visual evidence
evidence/live-ai-20261029.tap and .junit.xml	Actual provider replies and smoke results
evidence/frontend-checks.json and matching logs	Frontend unit tests, lint, typecheck and build
evidence/accessibility-observations.json	Post-fix mobile scan; zero sampled violations
evidence/database-cleanup-verification.json	Zero remaining temporary test schemas
evidence/source-baseline.json	Commit, working-tree state and non-secret source hashes
artifact-manifest.json	SHA-256 checksums for the packaged evidence

The package excludes environment files, API keys, database data directories, synthetic session fixtures and temporary browser traces. Reproduction uses the repository test scripts with a separately configured local MySQL account that can create/drop isolated test schemas. Run the backend coverage runner, frontend checks and Playwright suite separately, then verify database cleanup. Live AI tests require explicit opt-in configuration.

The browser runner emitted two internal step-reporting warnings during passing scenarios. All 13 assertions completed with no unexpected or flaky result; the JSON report retains the recorded outcomes. These warnings concern the test harness and remain distinct from application findings.